Live Prototype:
https://returlooptesting-khwd.vercel.app/

Checkpoint 6 · Usability Testing & Iteration Matrix

Empirical Testing Across 6 Campus Users, Quantitative Metrics & 4 Core Engineering Fixes | INT335

1. Quantitative Usability Testing Protocol & Results

Test #	Participant Name	User Profile	Task Evaluated	Duration	Task Success	Observed Usability Hitch
T1	**Rohan Verma**	Hosteller (2nd Yr CSE)	Report lost calculator with secret proof	48s	Passed	Confused between public description and secret proof.
T2	**Aman Sharma**	Day Scholar (3rd Yr Mech)	Browse campus found feed on mobile	30s	Passed	Filter dropdowns were too small on touch screen.
T3	**Officer Gurpreet**	Security Guard (Block 34)	Verify claim & release item	65s	Passed (Aid)	Struggled with technical English terms; needed Punjabi.
T4	**Simranjeet Kaur**	Fresher Hosteller (GH-2)	Review match alert notification	22s	Passed	Expressed fear that finder might see her phone number.
T5	**Staff Manjit Kumar**	Library Desk Supervisor	Log batch of 3 lost glasses	95s	Passed	Requested fast duplicate item logging for exam weeks.
T6	**Sahil Gupta**	iPhone User (Mobile Chrome)	Change language in navigation bar	Failed	Failed	Language modal was cut off at top under Dynamic Island.

2. Feedback-to-Iteration Matrix (The 4 Critical Engineering Fixes)

Hitch Identified	Root Cause Diagnostics	Engineering Fix Implemented in Code	Empirical Re-Test Validation
Fix 1: Mobile Modal Clipping	iOS WebKit viewport safe-area inset clipped modal inside sticky navbar stacking context.	Refactored `LanguageSelectorModal.jsx` using React `createPortal` rendering to `document.body` with `env(safe-area-inset-top)` padding.	Sahil Gupta re-tested on iPhone Chrome; modal rendered centered with 100% visibility.
Fix 2: Staff Language Barrier	Security guards found English technical terms intimidating, causing slow claim handovers.	Implemented complete 7 Indian regional languages localization (Hindi, Punjabi, Telugu, Tamil, Bengali, Marathi, English).	Officer Gurpreet completed handover verification in Punjabi in 18 seconds.
Fix 3: Secret Proof Clarity	Students accidentally typed public descriptions into the private proof field.	Added explanatory tooltips with a lock icon: 'Secret Proof: Only campus security can view this to verify your ownership.'	100% of participants entered private proofs (serial numbers, engraving) without confusion.
Fix 4: Vite Import Build Error	Missing `TranslatedText` component triggered build failure in `NotificationBell.jsx`.	Created unified `TranslatedText.jsx` wrapper component with robust fallback handling.	Vite production build compiled cleanly with 0 errors (`vite build` passed).

3. Empirical Usability Testing & Iteration Photographic Evidence

Photographic documentation of Team 15 conducting interactive usability testing sessions on campus, gathering live user feedback on the ReturnLoop interface and iterating on UI controls.

Testing Session 1: Indoor Task Walkthrough & Workflow Verification	Testing Session 2: Form Input Evaluation & Language Feedback



Team member evaluating user reaction to reporting flow and private proof clarification.



Participant reviewing multi-language selector and verifying ease of navigation.

Testing Session 3: Campus Outdoor Usability Test (Block Corridors)



Testing Session 4: Turnstile / Bin Transit Area Retrieval Review



Testing mobile responsiveness and quick-reporting under natural campus conditions.	Discussing physical drop-off and safety protocols near high-traffic campus zones.